



॥ वसुधैव कुटुम्बकम् ॥

Symbiosis

Institute of technology

(Established under section 3 of the UGC Act 1956)
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Semester: 3

Year: AY 25-29

ASSIGNMENT No: 2

TITLE: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management

Problem Statement: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

// DEFAULT CONSTRUCTOR

PROGRAM USING DEFAULT CONSTRUCTOR :

```
class Student {
```

```
// Instance variables

int rollNo;

String name;


// Default Constructor

Student() {

    rollNo = 101;

    name = "Sourabh Gupta";

}


// Method to display student details

void display() {

    System.out.println("Roll Number : " + rollNo);

    System.out.println("Name      : " + name);

}


// Main method

public static void main(String[] args) {

    Student s1 = new Student();

    s1.display();

}

}
```

Output :

```
C:\Users\HP\Documents\java>javac Student.java
```

```
C:\Users\HP\Documents\java>java Student
```

```
Roll Number : 101
```

```
Name       : Sourabh Gupta
```

```
C:\Users\HP\Documents\java>|
```

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// Parameterized Constructor

PROGRAM USING parameterized CONSTRUCTOR:

```
class Student {
```

```
    // Instance variables
```

```
    int rollNo;
```

```
    String name;
```

```
    // Parameterized Constructor
```

```
    Student(int r, String n) {
```

```
        rollNo = r;
```

```
        name = n;
```

```
    }
```

```
    // Method to display student details
```

```
    void display() {
```

```
        System.out.println("Roll Number : " + rollNo);
```

```
        System.out.println("Name       : " + name);
```

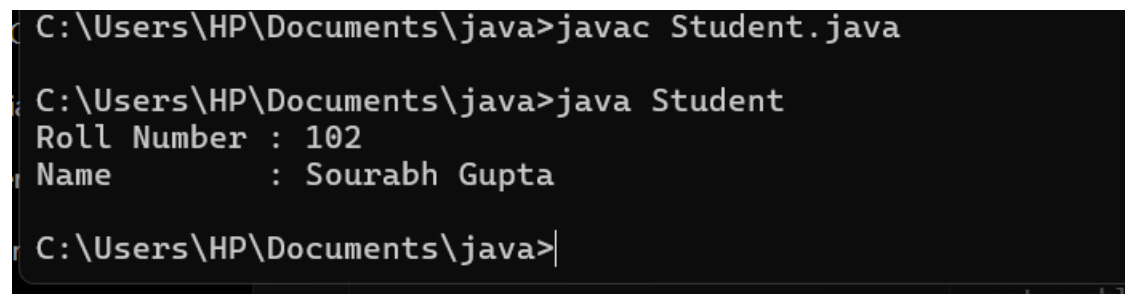
```
    }
```

```
    // Main method
```

```
    public static void main(String[] args) {
```

```
Student s1 = new Student(102, "Sourabh Gupta");  
s1.display();  
}  
}
```

Output :



```
C:\Users\HP\Documents\java>javac Student.java  
  
C:\Users\HP\Documents\java>java Student  
Roll Number : 102  
Name : Sourabh Gupta  
  
C:\Users\HP\Documents\java>|
```

// copy constructor

PROGRAM USING COPY CONSTRUCTOR:

```
class Student {  
  
    // Instance variables  
    int rollNo;  
    String name;  
  
    // Parameterized Constructor  
    Student(int r, String n) {  
        rollNo = r;  
        name = n;  
    }  
  
    // Copy Constructor
```

```
Student(Student s) {  
    rollNo = s.rollNo;  
    name = s.name;  
}  
  
// Method to display student details  
void display() {  
    System.out.println("Roll Number : " + rollNo);  
    System.out.println("Name      : " + name);  
}  
  
// Main method  
public static void main(String[] args) {  
  
    Student s1 = new Student(103, "Sourabh Gupta");  
  
    Student s2 = new Student(s1);  
  
    s2.display();  
}  
}
```

Ouput :

```
C:\Users\HP\Documents\java>javac Student.java  
C:\Users\HP\Documents\java>java Student  
Roll Number : 103  
Name       : Sourabh Gupta
```

EXERCISE 2.1 :

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

SOURCE CODE –

```
class Student {  
  
    // Instance Variables  
    String name;  
    int rollNo;  
  
    // Default Constructor  
    Student() {  
        name = "Sourabh Gupta";  
        rollNo = 101;  
    }  
  
    // Parameterized Constructor  
    Student(String name, int rollNo) {  
        this.name = name;  
        this.rollNo = rollNo;  
    }  
  
    // Method to display student details  
    void display() {  
        System.out.println("Student Name : " + name);  
        System.out.println("Roll Number : " + rollNo);  
        System.out.println();  
    }  
  
    // Main Method  
    public static void main(String[] args) {  
  
        System.out.println("Default Constructor");  
        Student s1 = new Student();  
        s1.display();  
  
        System.out.println("Parameterized Constructor");  
        Student s2 = new Student("Rahul Sharma", 102);  
        s2.display();  
    }  
}
```

OUTPUT -

```
C:\Users\HP\Documents\java>javac Student.java

C:\Users\HP\Documents\java>java Student
Default Constructor
Student Name : Sourabh Gupta
Roll Number : 101

Parameterized Constructor
Student Name : Rahul Sharma
Roll Number : 102

C:\Users\HP\Documents\java>|
```

EXERCISE 2.2 Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

PROGRAM –

class Mobile {

// Instance Variables

int mobileId;

String brand;

double price;

// Default Constructor

Mobile() {

mobileId = 101;

brand = "Samsung";

price = 24999.0;

}

// Parameterized Constructor

Mobile(int mobileId, String brand, double price) {

```
    this.mobileId = mobileId;
    this.brand = brand;
    this.price = price;
}
```

// Copy Constructor

```
Mobile(Mobile m) {
    mobileId = m.mobileId;
    brand = m.brand;
    price = m.price;
}
```

// Method to display mobile details

```
void display() {
    System.out.println("Mobile ID : " + mobileId);
    System.out.println("Brand    : " + brand);
    System.out.println("Price   : " + price);
    System.out.println();
}
```

// Main Method

```
public static void main(String[] args) {
```

// Default Constructor

```
    System.out.println("Default Constructor");
    Mobile m1 = new Mobile();
    m1.display();
}
```



```

        // Parameterized Constructor

        System.out.println("Parameterized Constructor");

        Mobile m2 = new Mobile(102, "OnePlus", 32999.0);

        m2.display();


        // Copy Constructor

        System.out.println("Copy Constructor");

        Mobile m3 = new Mobile(m2);

        m3.display();

    }

}

```

Output :

```

C:\Users\HP\Documents\java>javac Mobile.java

C:\Users\HP\Documents\java>java Mobile
Default Constructor
Mobile ID : 101
Brand      : Samsung
Price      : 24999.0

Parameterized Constructor
Mobile ID : 102
Brand      : OnePlus
Price      : 32999.0

Copy Constructor
Mobile ID : 102
Brand      : OnePlus
Price      : 32999.0

```

Conclusion : Thus, the program successfully demonstrated the use of default, parameterized, and copy constructors in Java. Constructors were used to initialize object data efficiently, while the copy constructor created duplicate object records. This assignment helped in understanding object initialization, data management, and the importance of constructors in object-oriented programming.